



Experiment-6

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1. Aim/Overview of the practical:

To implement and understand **VLAN (Virtual Local Area Network)** and **VLAN Trunking using IEEE 802.1Q**, and to verify communication between devices in the same VLAN and across multiple switches using **Cisco Packet Tracer**..

2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

3. Theory:

A) What is a Layer 2 Switch?

A Layer 2 switch is a network device that operates primarily at the Data Link Layer (Layer 2) of the OSI model. It uses MAC addresses to forward Ethernet frames and maintains a MAC Address Table to identify which device is connected to which switch port. The Cisco 2960 switch commonly used in Packet Tracer is primarily a Layer 2 switch.

B) What is a Layer 3 Switch?

A Layer 3 switch performs both Layer 2 switching and Layer 3 routing. Therefore, it can use both MAC addresses and IP addresses. It can also perform Inter-VLAN Routing, allowing devices in different VLANs to communicate when routing is configured.

C) VLAN (Virtual Local Area Network)

VLAN stands for Virtual Local Area Network. It allows one physical switch to be logically divided into multiple separate networks. Devices belonging to different VLANs are logically separated and placed in different broadcast domains.

For example:

VLAN 10 → STUDENTS

VLAN 20 → FACULTY

Devices in the same VLAN can communicate at Layer 2, while communication between different VLANs requires a router or Layer 3 switch with Inter-VLAN Routing configured. VLANs also reduce unnecessary broadcast traffic, improve security, simplify network management, and allow logical grouping of devices regardless of their physical location.

D) Access Port

An access port is a switch port that normally belongs to one VLAN and is generally used to connect end devices such as PCs, printers, and servers.

Example:

PC1 → Fa0/1 → VLAN 10

PC2 → Fa0/2 → VLAN 20

An access port carries traffic for the VLAN assigned to that port.

E) Trunk Port

A trunk port is a switch port that carries traffic from multiple VLANs over a single physical link. It is commonly used for connections between network devices, such as Switch-to-Switch connections.

For example, one trunk link can carry traffic for VLAN 10 and VLAN 20 between two switches.

F) VLAN Tagging and IEEE 802.1Q

When traffic from multiple VLANs travels over a trunk, the receiving switch must identify the VLAN to which each frame belongs. This is achieved through VLAN tagging.

The commonly used standard is IEEE 802.1Q, which carries VLAN information in Ethernet frames across a trunk link. The receiving switch uses the VLAN ID to identify the appropriate VLAN.

G) VLAN Trunking

VLAN trunking is the process of carrying traffic from multiple VLANs over a single physical network link between network devices.

4. Steps for Experiment/Practical:

A) VLAN Configuration Using One Switch

Procedure / Steps:

1. **Step 1:** Open **Cisco Packet Tracer** and place **one Cisco 2960 switch** and **four PCs** in the workspace.
2. **Step 2:** Connect all PCs to the switch using **Copper Straight-Through cables** according to the topology.
3. **Step 3:** Assign the IP addresses to the PCs:
PC1 → **192.168.10.11**
PC2 → **192.168.10.12**
PC3 → **192.168.20.11**
PC4 → **192.168.20.12**

Assign the subnet mask **255.255.255.0** to all PCs.

4. **Step 4:** Configure the VLANs on the switch.

Open the switch CLI and enter:

```
enable
configure terminal
vlan 10
name STUDENTS
exit
```

```
vlan 20
name FACULTY
exit
```

5. **Step 5:** Configure the ports connected to the VLAN 10 devices:

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit
```



```
interface fa0/2
switchport mode access
switchport access vlan 10
exit
```

6. **Step 6:** Configure the ports connected to the VLAN 20 devices:

```
interface fa0/3
switchport mode access
switchport access vlan 20
exit
```

```
interface fa0/4
switchport mode access
switchport access vlan 20
exit
```

7. **Step 7:** Verify the VLAN configuration using:

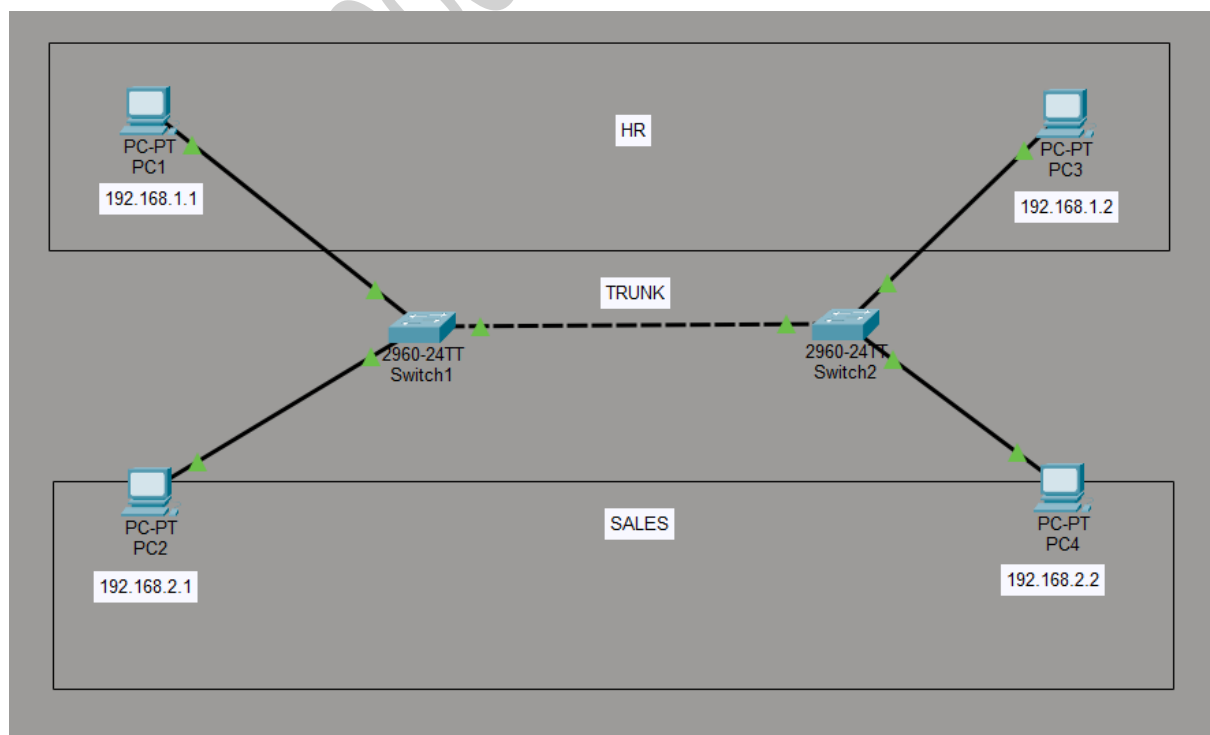
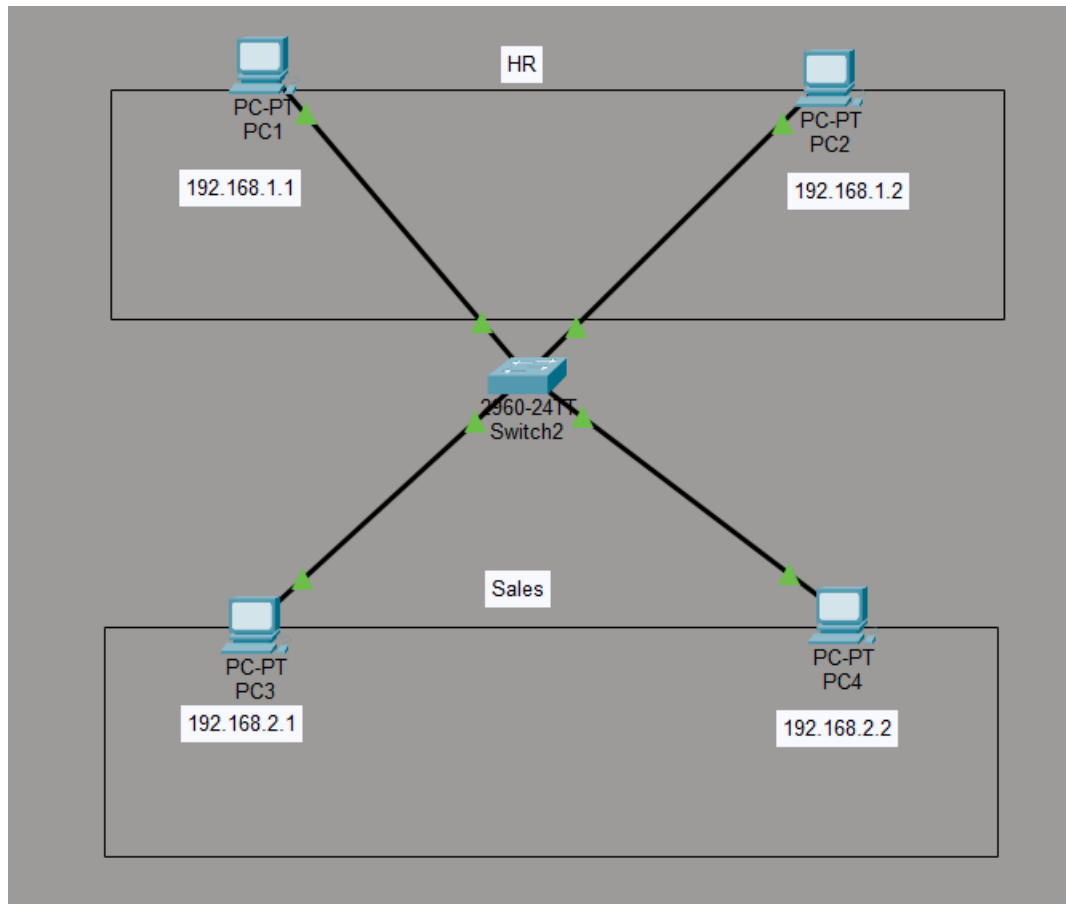
```
show vlan brief
```

B) VLAN Trunking Using Two Switches

Procedure / Steps:

1. Open **Cisco Packet Tracer** and place **8 PCs, 4 Routers, and 4 Switches** in the workspace.
2. Open Cisco Packet Tracer and place 4 PCs and 2 Cisco 2960 switches in the workspace.
3. Connect PC1 and PC2 to SW1 and PC3 and PC4 to SW2. Connect both switches using their Gi0/1 interfaces.
4. Assign IP addresses to the PCs according to their respective VLANs:
PC1 and PC3 → VLAN 10
PC2 and PC4 → VLAN 20
5. Create VLAN 10 and VLAN 20 on both switches.
6. Configure the PC ports as access ports and assign them to the appropriate VLAN.
7. Configure the link between SW1 and SW2 as a trunk port using:
switchport mode trunk
8. Verify the VLAN and trunk configurations using:
show vlan brief
show interfaces trunk

5. Output:



6. Result

The implementation of **VLAN and VLAN Trunking** was studied successfully using **Cisco Packet Tracer**. VLAN 10 and VLAN 20 were created and configured successfully on the switches. Communication between devices belonging to the same VLAN was verified successfully using the ping command. VLAN trunking was also configured between two switches, allowing multiple VLANs to communicate across a single physical link. Communication between different VLANs was not possible because **Inter-VLAN Routing** was not configured.

7. Learning Outcomes

- Understood the concept and importance of VLAN (Virtual Local Area Network).
- Learned how VLANs logically separate devices and create separate broadcast domains.
- Understood the difference between access ports and trunk ports.
- Learned how to create and configure VLAN 10 and VLAN 20 on Cisco switches.
- Configured access ports for end devices and assigned them to appropriate VLANs.
- Understood the concept of VLAN trunking and IEEE 802.1Q tagging.
- Learned how to configure a trunk link between two switches.
- Verified VLAN configuration and trunk status using show vlan brief and show interfaces trunk.
- Tested connectivity between devices in the same and different VLANs using the ping command.